

```

CAAAGCTCTTGCTTTGACAATTTGGTCTTT - - - CAGAA-TACTATAAAT human
CAAAGTTGTAGTCTTGACAATTCTGCTCTTT - - - ACATA-AAATTGAA GC rabbit
CAAAATTGTAGCCTTGACAATTTGACTCCTTTTACAAGA-AAATATAGGA cow

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Sequence comparison of the globin gene for human, rabbit, and cow. Dashes represent bases that are not present in the gene. Note the large amount of homology, as well as the highlighted differences.

Homology and Computer Analysis

To study homologous sequences, researchers use computer programs, such as BLAST (Basic Local Alignment Search Tool), to compare a DNA or protein sequence with a collection of other sequences. One such collection is GenBank, the genetic sequence database operated by the NCBI that contains all publicly available DNA and protein sequences. Biologists use databases such as GenBank to find out if a test sequence matches any known sequences, how well it matches, and which portions of the sequence match.

Computer programs identify matching sequences by similarity. However, similar sequences are not always homologous, because they may not have a common origin. Although many sequences that show similarity did evolve from a common ancestor, the appearance of similar sequences can also result from independent events. For example, mutations frequently occur in the gene for the envelope protein of the AIDS virus, HIV-1, changing the amino acid sequence of the protein. The human immune system recognizes and destroys unmutated viruses, while leaving unharmed (selecting for) those viruses that contain mutations that make them unrecognizable. As a result, viruses from different patients can show identical mutations in the envelope protein, even though the patients were infected by different strains of the virus.

Exploring the Mechanisms of Mutation

The ability to compare protein and DNA sequences not only shows us where evolution has occurred but provides insight into its mechanisms. By comparing genomes, we find that mutations can occur on a small scale: Even a single nucleotide change is a mutation. They can also occur on a large scale, as happens when sequences are inserted, deleted, duplicated, or moved between chromosomes.

Many mutations that replace single nucleotides have no effect because of the “degeneracy” or redundancy of the genetic code. The genetic code has more **codons** (sixty-four) than amino acids (twenty). As a consequence, most amino acids are specified by two to four different codons. Because of this, some mutations can be “silent,” with one nucleotide replacing another but without changing the specified amino acid. Other mutations are said to be “conservative.” This occurs when a mutation replaces one amino acid with another that has similar properties: They may be chemically similar sharing the same charge, shape, or polarity.

codons sequences of three mRNA nucleotides coding for one amino acid

If, however, the mutation affects the function of an important protein, that mutation may result in an evolutionary dead end, because it is less likely to be passed on to a future generation. As a result, important sequences show fewer mutations, whereas less important sequences show more change. Such properties can be deduced by comparing sequences from different organisms. Proteins that interact with other molecules, such as DNA or RNA, tolerate fewer changes in structure, and show little change



molecular systematics
the analysis of DNA and other molecules to determine evolutionary relationships

through evolution. The histone proteins that form the backbone of the eukaryotic chromosome are important examples.

Evolutionary Relatedness

The number and types of differences that accumulate between genes or proteins of two different species can be used to assess their evolutionary relatedness and the amount of time since they diverged from a common ancestor. Such studies, termed “**molecular systematics**,” can be used to show that humans are more closely related to chimps than to gorillas, for instance, and how long ago the split in these lineages occurred.

Homologous proteins that perform the same function in different species are called orthologs. For example, hemoglobin, a protein that transports oxygen, has a similar amino acid sequence in both horses and dogs. If the predicted amino acid sequence of a newly discovered protein is similar to a known protein in another species, researchers can make guesses about the function of the newly discovered gene. If the sequence of a newly discovered protein was similar to hemoglobin, one might guess that the new protein is able to bind to oxygen and function in transporting oxygen. In the way, orthologs help researchers about the functions of newly discovered genes.

Natural selection acts against harmful mutations in critical genes. Gene duplication, however, makes extra copies of less critical genes, which are more free to acquire mutations. Members of these gene families are known as paralogs. Researchers look for paralogs in order to find proteins with new abilities. Cytokine genes, for example, are all derived from the same ancestral gene and share common sequence motifs, yet they fill a variety of roles in the immune system. New members of the cytokine family might be valuable tools for fighting disease. Just as species diverge and fill new biological niches, genes become duplicated and acquire new functions. On a molecular scale, the evolution of the genome reflects the evolution of all living things. SEE ALSO BIOINFORMATICS; CHROMOSOME, EUKARYOTIC; EVOLUTION OF GENES; MOLECULAR ANTHROPOLOGY; MUTATION.

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Internet Resource

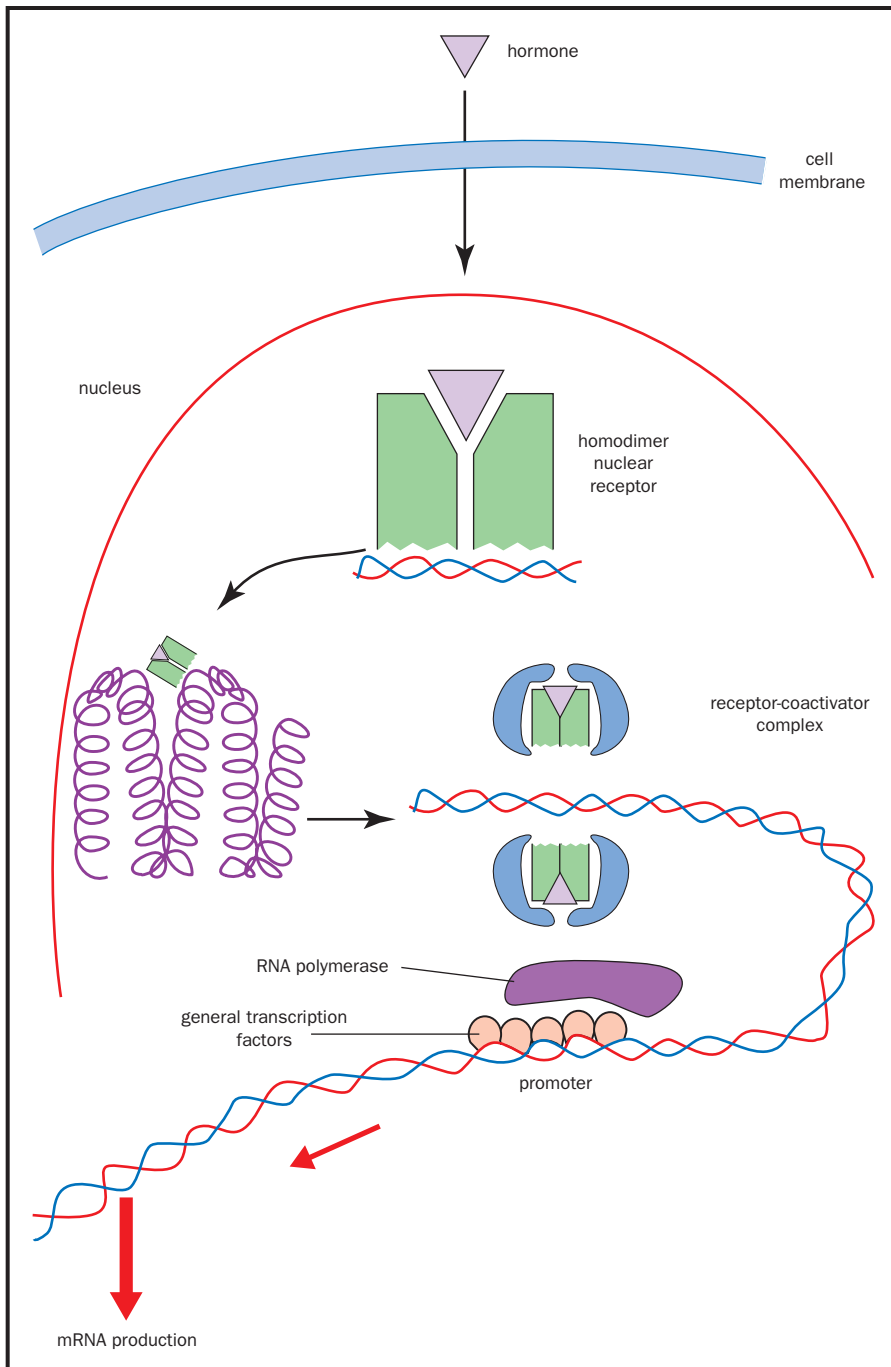
National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov>>.

chemotaxis movement of a cell stimulated by a chemical attractant or repellent

pheromone molecule released by one organism to influence another organism’s behavior

Hormonal Regulation

All types of cells are capable of receiving signals from their environment and mounting an appropriate response to the signal, such as **chemotaxis** toward a nutrient source or toward other cells emitting a **pheromone**. The



A steroid hormone binds to a two-part receptor within the cell. This links with a coactivator, making a complex that binds to DNA. This triggers transcription of the target gene, through interaction with general transcription factors.

key difference between microorganisms and more-complex plants and animals is that the former are largely independent, with each cell in contact with the environment. In contrast, more complex plants and animals are self-contained entities whose interior is mostly insulated from the environment. Animals have complex organ systems, with each organ specialized for a particular function. Therefore, the survival of the organism depends on the precise regulation of growth, differentiation, and metabolism in different groups of cells throughout the animal.

The endocrine system is the set of glands and other tissues responsible for coordinating cellular growth and differentiation, many aspects of

NUCLEAR RECEPTOR PARTNERS FOR RXR

Receptor	Hormone	Function
retinoic acid receptor (RAR)	all-trans retinoic acid (vitamin A derivative)	regulates important aspects of early embryonic development
thyroid hormone receptor (TR)	triiodothyronine	controls the body's basal metabolic rate
vitamin D3 receptor (VDR)	vitamin D3	regulates calcium homeostasis, other functions
fatty acid receptor (PPAR)	fatty acid and eicosanoid ligands	regulates fat metabolism and the body's ability to utilize insulin
bile acid receptor (FXR)	bile acid	regulates the transport of bile acids and cholesterol out of the cell
oxysterol receptor (LXR)	oxysterol	regulates the formation of bile acids from cholesterol
benzoate receptor (BXR)	unknown	unknown
steroid and xenobiotic receptor (SXR)	multiple compounds	regulates degradative and detoxification enzymes
constitutive androstane receptor (CAR)	multiple compounds	regulates degradative and detoxification enzymes

homeostasis maintenance of steady state within a living organism

reproduction and embryological development, the maintenance of **homeostasis**, and a variety of cyclical phenomena (e.g., reproductive cycles). Because these varied processes require coordinated gene expression, they are regulated by a large and diverse group of inter- and intracellular signaling pathways.

The primary mediators of these pathways are a large group of chemical messengers, called hormones, produced by specialized cells in response to physiological requirements. Many of these specialized cells are located in endocrine glands. Some of the most well-known endocrine glands are the pituitary (which produces many important hormones, such as ACTH), the thyroid (which produces thyroid hormone to regulate metabolic rate), the adrenals (which produce glucocorticoids to regulate blood sugar and stress, and which also produce epinephrine, or adrenaline), the testes (which produce testosterone), and the ovaries (which produce estrogens and progesterone).

Hormones may need to act at different distances from their source, depending on the requirements of the organism, and they can be broadly classified according to the distance across which they signal. Endocrine hormones (e.g., adrenocorticotrophic hormone) are produced by endocrine glands (in this case the pituitary gland, located at the base of the brain) at a distance from their site of action (in this case adrenal glands, sitting atop the kidneys) and must be transported throughout the body via the circulatory system. Paracrine hormones, such as the prostaglandins that mediate local inflammatory processes, are produced near their site of action. Autocrine hormones, such as interleukins, act on the cells that produce them, in this case the white blood cells of the immune system.

Hormone Receptors

Regardless of the distance across which a hormone acts, only those cells that contain a specific receptor can respond to the corresponding hormonal signal. The expression of receptors only in the target cells ensures that these (and only these) cells respond in the appropriate way to the hormone, despite the possible presence of a large number of other hormones in the immediate surroundings.

In addition to the distance across which they act, hormones may be further divided into two large groups based on where in the target cell the hormone receptors are located. The first class consists of extracellular hormones that act via specific cell-surface receptors. Most hormones of this family are proteins (such as insulin, interferons, interleukins, and growth factors), fatty acid derivatives (such as prostaglandins and leukotrienes), or amino acid derivatives (such as serotonin and melatonin).

Extracellular hormones bind to specific receptors on the cell surface, triggering a chain of events inside the cell. These events may include the modification (e.g., **phosphorylation** or dephosphorylation) of one or more “second messengers”—small molecules that act inside the cell to continue the signaling cascade. In addition to any other short-term effects they may have, virtually all hormonal signaling processes culminate in a change in the expression of a set of target genes. Depending on the cell, the transcription of these genes may be increased or decreased, or be turned completely on or turned off, in response to the presence of the hormone.

The other major class of hormonal signals consists of small, typically fat-soluble molecules that are able to diffuse freely into cells. Once inside the cell, the hormone binds to its receptor to directly regulate the expression of target genes. The hormone-receptor complex is a functional **transcription factor** that in most cases leads to the expression of its target genes. This modulation primarily occurs directly, via binding of the hormone-receptor complex to target DNA sequences, although additional regulation can occur indirectly, via interaction with other transcription factors.

Since the vast majority of these receptors are always in the nucleus, the family is referred to as the “nuclear hormone receptor superfamily.” It is also often also called the “steroid receptor superfamily,” because steroid receptors were the first of this family to be identified. Steroid hormones include testosterone, progesterone, and the estrogens. The discussion that follows will focus on hormones that interact with nuclear receptors to directly influence gene expression.

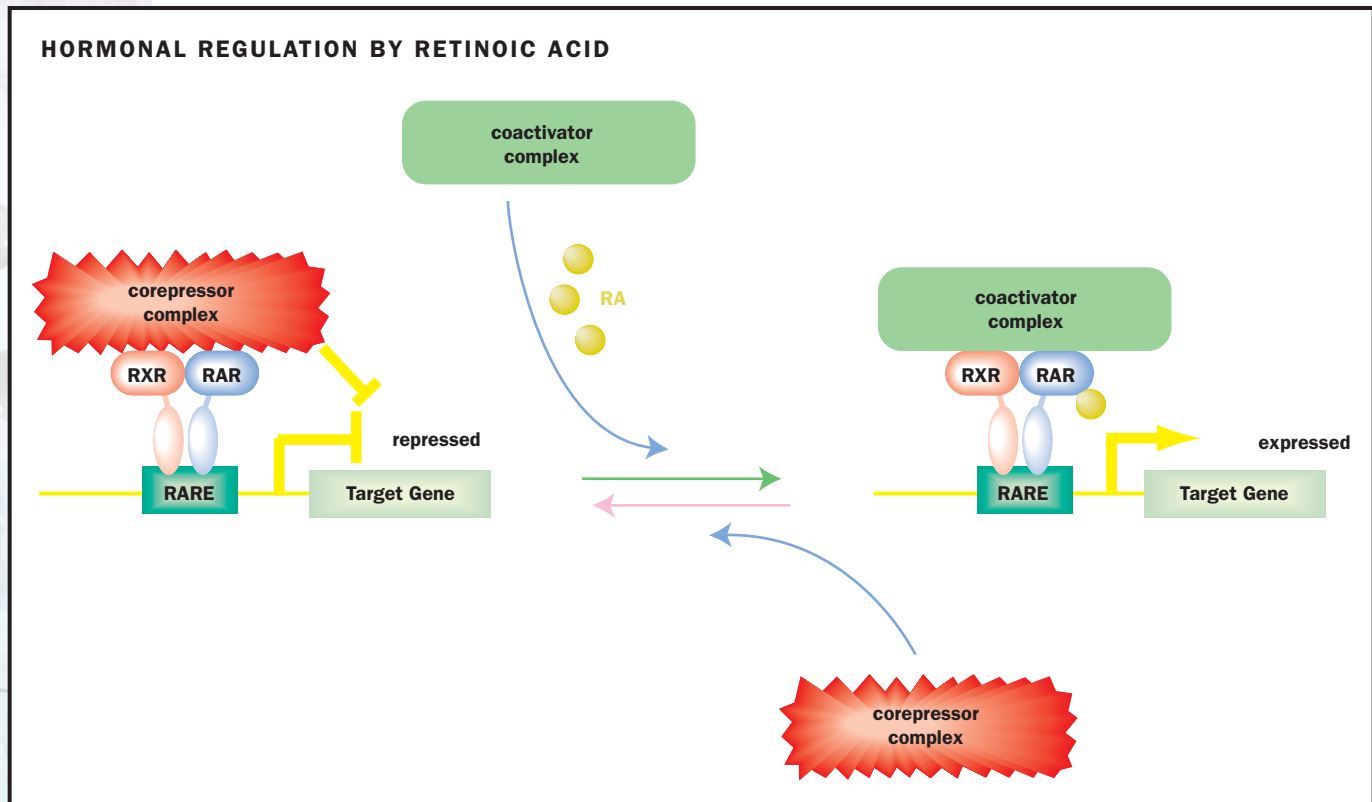
Nuclear Receptors and Their Hormones

The nuclear receptors are a large group of related proteins that mediate many of the effects of steroid hormones, thyroid hormone, vitamin D3, the vitamin A derivative retinoic acid, and modified forms of cholesterol, such as hydroxycholesterol and bile acids. The number of nuclear hormone receptor genes varies widely among animals. Humans and other vertebrates have about forty-nine receptor genes, whereas the nematode *Caenorhabditis elegans*, with only 959 cells in the adult worm, has more than 250 receptor genes. This was a somewhat unexpected finding, and it led to the speculation that

phosphorylation addition of the phosphate group PO_4^{3-}

transcription factor protein that increases the rate of transcription of a gene





Schematic depiction of how the retinoic acid receptor activates transcription of target genes. In the absence of retinoic acid, the receptor interacts with the corepressor complex to condense chromatin and silence transcription from target genes. In the presence of retinoic acid, the corepressor complex dissociates and the receptor can then interact with a coactivator complex that leads to chromatic decondensation and subsequent activation of target genes. The active and repressed states exist in a dynamic equilibrium that is regulated by the presence of retinoic acid.

C. elegans may use nuclear hormone receptors to regulate processes that are controlled by different transcription factors in vertebrates.

The classical steroid receptors are all quite similar to each other, and they all function as homodimers—a complex of two identical proteins. With the exception of the estrogen receptor, all of these receptor homodimers bind to exactly the same target DNA sequences. For this reason, high levels of one hormone may cause inappropriate activation of another pathway and multiple consequences.

It is currently unclear how one receptor (e.g., the glucocorticoid receptor) distinguishes its correct target genes from those of other receptors (e.g., the progesterone receptor), when multiple receptors are present in the same cell. Estrogens, progesterone, and androgens are important steroid hormones that influence many aspects of later development.

The estrogen receptor is expressed in the brain, kidney, liver, and lungs, and throughout the female reproductive tract. Interestingly, the estrogen receptor is also present and required in male reproductive tissues. The major human estrogen, 17- β estradiol, activates the receptor to regulate cell proliferation in, for example, the uterus. The progesterone receptor is also important for female development, with its effects restricted to the female reproductive tract and mammary tissue.

The androgen receptor is primarily responsible for male development and secondary sexual characteristics, such as muscle mass. The major hormone acting through this receptor is dihydrotestosterone (DHT). The androgen receptor is also the major receptor targeted by so-called **anabolic steroids**, which function by mimicking the activities of DHT on muscle growth. Some predictable and unfortunate consequences of increasing the circulating levels of testosterone-like molecules include atrophy of the testes (since they sense high levels of testosterone and react by shutting down their own production) and the development of female secondary sexual characteristics, such as breasts, in men (because excess DHT is converted to estradiol).

The largest and most diverse group of nuclear receptors contains those that function as heterodimers, meaning they are composed of two different parts. Each heterodimer is composed of one unique receptor protein and one protein common to the whole group, called the 9-cis-retinoic acid receptor (RXR). There are nine distinct hormone-regulated receptor-signaling pathways wherein RXR is used as a common heterodimeric partner.

One of these is the retinoic acid receptor (RAR), which binds with all-trans retinoic acid, a vitamin A derivative, to regulate many important aspects of early embryonic development, including limb formation, central nervous system patterning, growth and differentiation of many tissues, **hematopoiesis**, and eye, brain, and craniofacial development. Since retinoic acid affects so many important developmental processes, too much or too little retinoic acid has profound effects on early development.

Another RXR partner is the steroid and **xenobiotic** receptor (SXR). Steroid and xenobiotic **ligands** for SXR regulate the breakdown of foreign chemicals by degradative enzymes in the liver and intestines, protecting the body from toxic chemicals and bioactive dietary compounds. SXR is known to directly regulate the transcription of genes such as *CYP3A4*, which mediates the breakdown of 60 percent of clinically useful drugs, as well as the transcription of the multidrug resistance protein MDR1, which transports drugs out of the cell. Thus, SXR is a key mediator of the body's defense system against foreign chemicals, controlling both their metabolism and clearance from the cell.

Nuclear Hormone Receptors and Transcriptional Regulation

As noted above, nuclear receptor hormones generally act as transcription factors to increase transcription of their target genes. They do this by increasing the rate at which **RNA polymerase** binds to the target gene's **promoter**. This occurs in several steps.

The binding of a hormone to the receptor triggers the assembly of other proteins to form a "coactivator complex." The hormone-receptor-coactivator complex binds to a specific DNA sequence (called the hormone response element, a type of transcriptional enhancer). This complex then alters the local DNA structure by directly or indirectly chemically modifying the **histones**. These modifications open up the DNA, increasing access to the target genes and thereby allowing RNA polymerase and other (general) transcription factors to reach the gene promoter region. Additionally,

anabolic steroids hormones used to build muscle mass

hematopoiesis formation of the blood

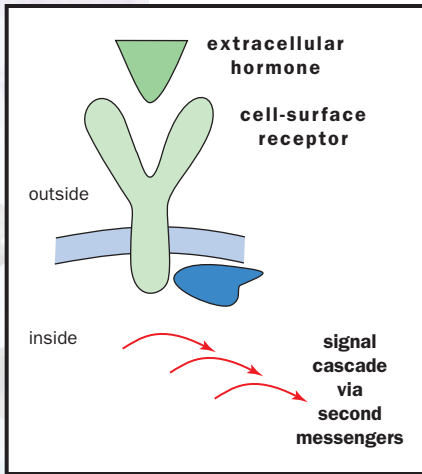
xenobiotic foreign biological molecule, especially a harmful one

ligands molecules that bind to receptors or other molecules

RNA polymerase enzyme complex that creates RNA from DNA template

promoter DNA sequence to which RNA polymerase binds to begin transcription

histones the proteins around which DNA wind in the chromosome



Some hormones trigger second-messenger signal cascades, quickly turning on cellular enzymes for a rapid response.

mRNA messenger RNA

dimerize linkage of two subunits

enzymes proteins that control a reaction in a cell

the hormone-receptor-coactivator complex can directly interact with general transcription factors to help form a “preinitiation complex” of proteins on the target gene promoter. RNA polymerase then interacts with this complex, and the transcription of the gene into **mRNA** begins.

All hormone-regulated nuclear receptors activate transcription in this manner. Some, such as the steroid receptors, exist in cells as cytoplasmic complexes with “chaperone proteins,” such as HSP90, and are excluded from the nucleus in the absence of the hormone. In the presence of hormone, the complexes dissociate, and the receptors **dimerize** and are transported to the nucleus, where they activate transcription.

Other receptors, such as the retinoic acid, thyroid hormone, or vitamin D receptors, are always found in the nucleus and interact with their specific target genes in the presence or absence of the hormone. When the hormone is absent, the receptor interacts with “corepressor” proteins. The complex of receptor and corepressor interacts with histone deacetylases, leading to local chromatin condensation and silencing of the target gene. Hormone binding leads to a change in the three-dimensional structure of the receptor, causing dissociation of the corepressor complex and leading to the recruitment of the coactivator complex, which enables the target gene to be transcribed.

The Importance of Hormone Concentration

Because the hormones that act through nuclear hormone receptors are nearly all fat-soluble, they are readily absorbed into the body, freely transported, and stored and accumulated in fatty tissues. Steroids, retinoic acid, thyroid hormone, and vitamin D3 are active at extremely low concentrations, ranging from about 0.3 to 30 parts per billion, with 3 parts per billion considered a physiological concentration for retinoic acid and many steroids. Since the hormones are present and act at such low concentrations, it is critical that their levels be precisely regulated. Consequently, hormone synthesis and degradation is regulated by the activity of specific biosynthetic and catabolic **enzymes**.

It should also be noted that some chemicals in the environment and natural compounds found in the diet can affect the activity of hormone receptors, particularly the estrogen receptor. Such interaction can potentially lead to disturbances in hormone homeostasis and inappropriate regulation of target genes. These xenobiotic “endocrine disrupting chemicals” have the potential to impact many body systems by inappropriately activating or interfering with the activity of hormone receptors. As a result, endocrine disruption is a growing concern that is being studied intensively in many laboratories around the world. SEE ALSO CHAPERONES; ROUNDWORM: *CAENORHABDITIS ELEGANS*; SIGNAL TRANSDUCTION; TRANSCRIPTION FACTORS.

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HPLC: High-Performance Liquid Chromatography

High-performance liquid chromatography (HPLC) is an advanced form of liquid chromatography used in separating the complex mixture of molecules encountered in chemical and biological systems, in order to understand better the role of individual molecules. In liquid chromatography, a mixture of molecules dissolved in a solution (mobile phase) is separated into its constituent parts by passing through a column of tightly packed solid particles (stationary phase). The separation occurs because each component in the mixture interacts differently with the stationary phase. Molecules that interact strongly with the stationary phase will move slowly through the column, while the molecules that interact less strongly will move rapidly through the column. This differential rate of migration facilitates the separation of the molecules.

The advantages of HPLC over other forms of liquid chromatography are several. It allows analysis to be done in a shorter time and achieves a higher degree of resolution, that is, the separation of constituents is more complete. In addition, it allows stationary columns to be reused a number of times without requiring that they be regenerated, and the results of analysis are more highly reproducible. A further advantage of HPLC is that it permits both instrumentation and quantitation to be automated.

Components of HPLC Analysis

HPLC has four basic components: a solvent delivery system to provide the driving force for the mobile phase; a means by which samples can be introduced into the solvent; the column; and some type of detector. A recorder is used to display the results and an integrator performs the calculations.

The column used for a specific separation is based on the type of the molecules to be analyzed. Various types of chromatographic modes can be used for the separation of the molecules. For example, ion exchange columns separate charged molecules such as amino acids, proteins, or **nucleotides**. Size exclusion columns separate organic polymers such as polyvinyls and silicones or biopolymers such as proteins, nucleic acids, or sugars. Adsorption columns separate molecules based on their interaction with the stationary phase. This mode is useful for the separation of vitamins, dyes, lipids, phenols, and antioxidants. Partition columns are used to separate molecules based on the way that the solvent becomes partitioned into stationary and mobile layers, and is useful in analyzing steroids, aromatics, vitamins, and antibiotics. The molecules **eluting** from any one of these different types of column are then analyzed by various types of detectors, measuring absorbance, fluorescence, or electrochemical or radiochemical properties. Other types of detectors include mass spectroscopy and refractive index.

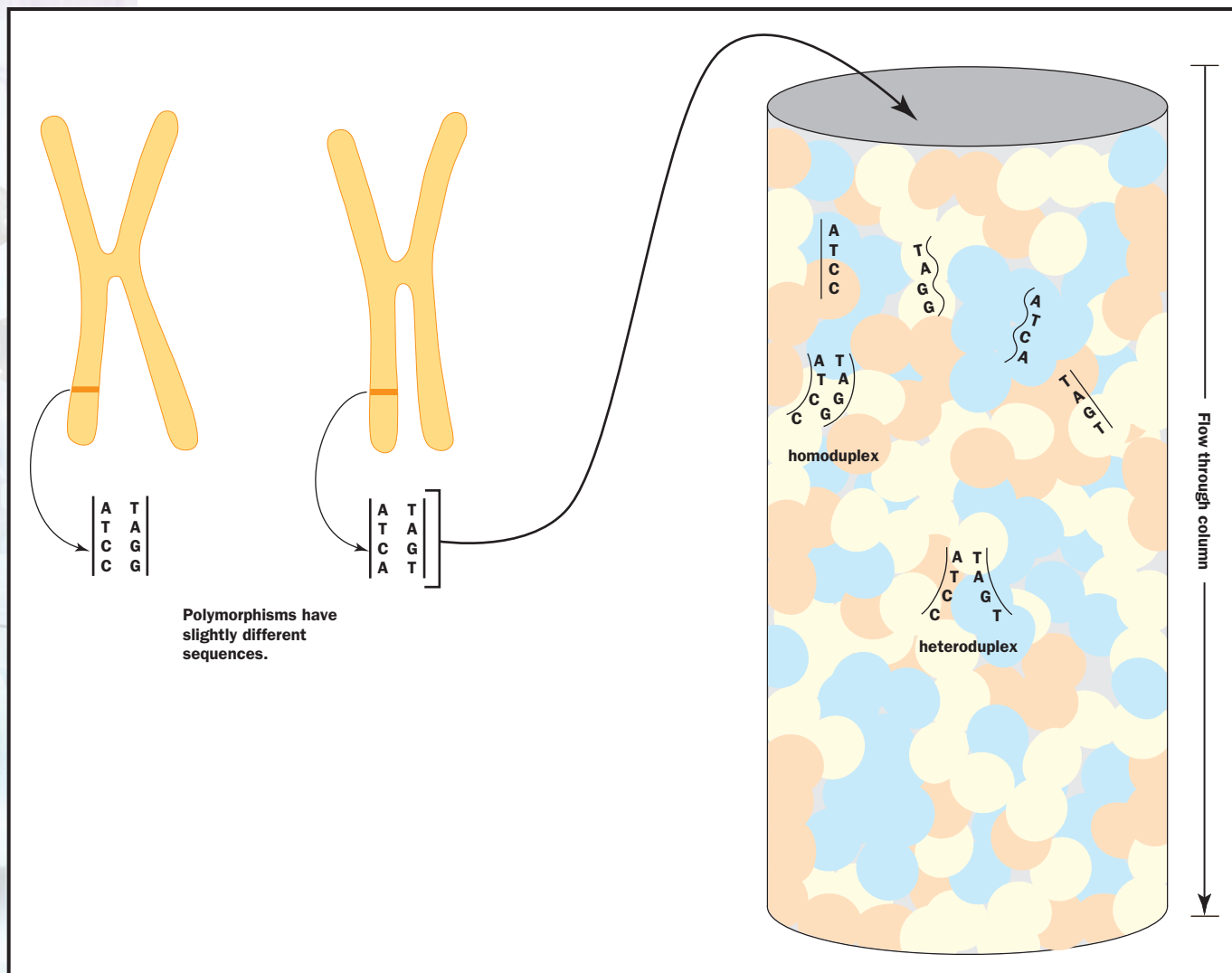
HPLC Applications

A recent advancement of HPLC has been the development of the denaturing HPLC method (DHPLC). This procedure can separate

nucleotide the building block of RNA or DNA

eluting exiting





DNA from different sources is placed into the column and treated to allow it to separate and recombine. Two strands from the same source, termed a homoduplex, bind strongly and tend to remain together longer. Heteroduplexes do not attract as strongly, and remain unattached more often.

base pair two nucleotides (either DNA or RNA) linked by weak bonds

polymorphisms occurring in several forms

double-stranded DNA molecules that differ by as little as one **base pair**. The speed of analysis (approximately 5 minutes per sample) and the size of DNA fragment that can be analyzed (up to 2.0 kilobytes) has made it a preferred method for a variety of applications in the field of molecular biology. Applications of DHPLC include the detection of single nucleotide **polymorphisms** (SNPs). These are single base-pair variations in DNA that can give valuable information on genetic variation within a population. They can also help to identify the genes that cause certain human diseases.

To determine if the two genes of interest differ, they are first amplified by the polymerase chain reaction and then injected together into a so-called reversed-phase column. In this type of column, the stationary phase is less polar than the mobile phase, which is the opposite of the arrangement found in standard columns. Once the genes are injected, their DNAs bind to the stationary phase. Increasing the temperature causes each gene to separate

into its two strands (a phenomenon called denaturing). Once denatured, the strands can enter the mobile phase and move through the column.

Cooling the column causes the strands of the genes to rejoin, and the DNAs reattach to the column. The temperature is manipulated to make the strands constantly separate and rejoin, with the balance determined by the strength of the attraction that exists between the strands. If the two genes are exactly identical, they will spend more time in the stationary phase, and elute from the column more slowly. If the genes differ even by a single nucleotide, however, they will spend more time in the mobile phase, and leave the column more quickly.

Y chromosome analysis is one of the most powerful molecular tools for tracing human evolution. Polymorphisms in the human Y chromosome, detected by DHPLC, can be used as markers for tracing human evolution. This will eventually help to elucidate the patterns of human origins, migration, and mixture. The ability to rapidly and efficiently **genotype** SNPs by use of DHPLC is also useful in medicine, through the identification of mutations that result in susceptibility to certain diseases or that affect physical responses to certain drugs. SEE ALSO DNA; POLYMERASE CHAIN REACTION; POLYMORPHISMS; Y CHROMOSOME.

Prema Rapuri

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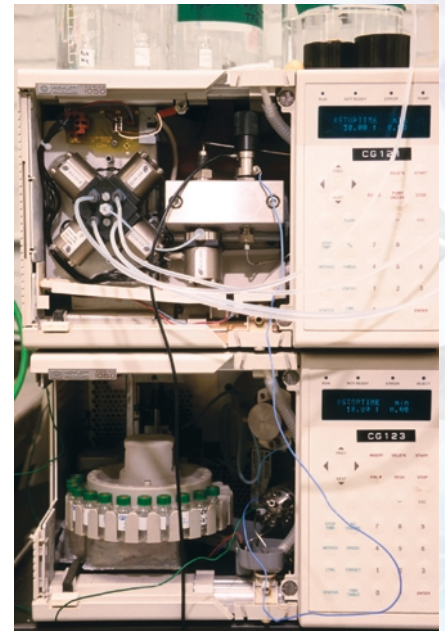
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Human Disease Genes, Identification of

In order to help treat human diseases, it is important to understand what causes them to occur. Understanding what causes a disease is the first step in understanding the entire abnormal course of disease. Sometimes it is fairly easy to determine what causes a disease. For example, pneumonia is caused by the *Pneumococcus* bacterium. However, in other cases it is not nearly as easy to tell what is causing a disease, so scientists look for clues from a number of different sources.

One such clue is having a disease run in families, which suggests that a disease might be caused by a gene or genes passed from parent to child, perhaps over many different generations. The process of identifying these genes, called disease gene discovery, is important because it helps scientists to understand what is going wrong as a result of such diseases, called the disease pathogenesis. By understanding the disease process, it is possible to figure out where it is easiest to either stop or correct what is going wrong. Variations or mutations within a gene may themselves act to cause the disease. Alternatively, they may modify the risk of developing the disease or modify how it is expressed (i.e., what the symptoms are). Thus genes may be important both directly and indirectly in causing disease.



A high-performance liquid chromatography system allows geneticists to perform molecular research efficiently by automating necessary processes.

genotype set of genes present



SOME EARLY DISEASE-CAUSING GENES FOUND USING GENOMIC SCREENING

Disease name	Symptoms	Year identified	Chromosome
Chronic Granulomatous Disease	Poor immune system	1986	X
Duchenne Muscular Dystrophy	Muscle weakness, muscle deterioration	1986	X
Cystic Fibrosis	Difficulty breathing, poor lung function	1989	7
Neurofibromatosis	Benign tumors	1990	17
Fragile-X Syndrome	Mental retardation	1991	X
Huntington's Disease	Uncontrollable movements, brain deterioration	1993	4
Tuberous Sclerosis	Benign tumors	1993	16
Alzheimer's Disease	Loss of memory and brain function	1993	19
Breast Cancer	Tumors in the breast	1994	17
Glaucoma	Loss of eyesight	1997	1

Table 1.

The Process of Disease-Gene Discovery

Performing disease-gene discovery requires asking and answering five important questions: What does the disease actually look like? Do genes really play an important role in this disease? Can enough families with enough affected members be found to help study the disease? What are the genes? What do these genes do?

What does the disease actually look like? It may seem obvious that to study the genetics of a disease such as epilepsy (seizures), families with seizure sufferers should be studied. However, it is possible that genes are important only in some, not all, types of epilepsy. It may also be true that different genes are important in different types of epilepsy. Thus it may be better not to study all seizures but only one type. For example, it might be better to study seizures that affect only one part of the body or that cause only a loss of consciousness with no effects on the rest of the body. Which type of seizures to study depends mostly on how important genes are for that particular type of disease.

Do genes really play an important role in this disease? There are many ways to see if genes are important in a disease (or a particular subtype of disease) without having to know what the genes are. Figuring this out requires studies that look at information from very large collections of families and individuals. These are usually called **population studies**.

population studies collection and analysis of data from large numbers of people in a population, possibly including related individuals

One important type of population study looks at a large set of twins to see how often two identical twins (who have all the same genes) both have a disease and compares that to how often two fraternal twins (who are just like brothers and sisters in that they have only half of their genes in common) both have the disease. If genes are important, the identical twins will share the disease (be concordant) much more often than fraternal twins. For example, in over 80 percent of cases where one identical twin is diagnosed with autism, the other is also diagnosed with the disorder. When one fraternal twin, on the other hand, has autism, the other will have the disorder in only about 5 percent of cases. This suggests that genes have a very strong effect in autism.

Another way to see if genes are important is to look at a large set of adopted children who have the disease and compare how often their adoptive parents have the same disease with how often their natural parents have it. If the rate of the disease is much higher in the natural parents than in the adoptive parents, genes are likely to be important. For example, the fre-

quency of multiple sclerosis in the natural parents of people with multiple sclerosis is about 3 percent. Among the adoptive parents of people with multiple sclerosis, the frequency was much smaller and about the same as the general population.

Yet another way to see if genes are important is to look at the occurrence of the disease in the brothers and sisters of someone who has the disease. If the rate at which brothers and sisters have the disease is much higher than the rate in the overall population, then it is likely that genes are important. Autism again is a good example. The rate at which autism is found in the brothers and sisters of an affected person is about 3 in 100, which does not seem very high. But the rate in the general population is only about one in five hundred (0.2%). It is the comparison of these rates that is important, not just the actual frequencies.

Each of these kinds of studies requires looking at a lot of people and their families, and each will usually take several years to complete. However, such studies are very important, since there is no point in looking for genes that affect a disease if we know that genes are not important. It is only after this information is available and genes are known to be important that the next step can be taken.

Can enough families with enough affected members be found to help study the disease? Depending on the disease, the type of family that can be found for genetic studies may differ a lot. For diseases caused by a **mutation** in a single gene, families with many affected people can be found. Sometimes these families may have as many as twenty or thirty affected people, over three or four generations. These types of families are usually quite rare, and thus it can take a lot of work to find them. On the other hand, having even a single large family can be enough to allow a gene to be found.

For diseases where genes may have only a moderate effect, smaller families, where only two or three people (usually brothers and sisters) are affected, may be the only ones that can be found. Since such diseases tend to be more common in the general population, it is easier to find these families than to find the very large families. However, the smaller families cannot, by themselves, help much in finding the location of a disease gene. Therefore many such families (usually hundreds) are needed. How the actual process of finding the genes is done depends on the type of families that are studied.

One of the important aspects of finding the families is deciding how to ask them to be part of the study. It is important that each person who participates is told what he is going to have to do. In most cases, participants will just be answering a lot of questions, giving permission to get medical records about their disease, and giving a blood sample. Sometimes additional hospital tests might be needed. Before they can be studied, potential subjects must be asked to participate and must give “**informed consent**,” which simply means that they have been told what they will need to do, what the risks (if any) are, and have agreed that this is fine with them. It is one of the overriding rules of human genetic studies that each person must volunteer for the study, that he gives informed consent, that he has the right to refuse, without the refusal affecting his medical care, and that he can withdraw from the study at any time.

What are the genes? Once the information and blood samples are collected from the families, the process of finding the genes can begin. There

mutation change in
DNA sequence

informed consent
knowledge of risks
involved



are two ways to search for them. The first involves looking across all the chromosomes, using genetic maps, and trying to correlate the occurrence of the disease in a family with the occurrence of one or more genetic markers from the genetic maps. This approach is called genomic screening and tries to find the genes based only on their location. It does not require that anything be known about what any of the genes do. Genomic screening uses a number of special statistical techniques to look at the probability that the disease gene (whose location is unknown) and one or more of the markers (whose location is known) are located near each other. One difficulty is that the location will not be known precisely. That is, this method will point to a region that may contain as many as 500 genes. This is a lot less than the 50,000 or so thought to exist in the human genome, but it is still a lot of genes that need to be tested. Genomic screening has worked spectacularly well for hundreds of diseases where there is a single causative mutation in a gene. A list of some of these genes is given in Table 1.

The second approach is to look at one or more specific genes to see if they might be directly involved in the disease. This is called the candidate gene approach. A gene becomes a candidate when something is known about its function and when this function might have something to do with the disease. For example, if a gene was involved in the development of the cornea of the eye, it would be a good candidate for any disease that affects the development of the cornea. The success of the candidate gene approach depends on two things. The first is how much is known about the disease process, and the second is how much is known about the function of the genes.

It is also possible to combine the two approaches. The genomic screening approach may identify several regions on several chromosomes that might contain a disease gene, but these regions may contain hundreds of genes each. By looking at the functions of these genes, it may be possible to identify one or just a few that are the most likely to be involved in the disease. These genes can then be tested using the candidate gene approach.

What do these genes do? Once a gene has been identified as being involved in the disease, it is important to further study the gene to find out what it does under normal circumstances, and what it is doing when it is changed and causing disease. Sometimes a lot is known about the normal function, but many times very little is known. Studies to look at the function may include studies of the normal gene in living cells that are grown in the laboratory. Another type of study involves testing the same normal genes in other living organisms such as mice, rats, and fruit flies. Animal studies are very helpful because animals can be tested in ways not possible in humans. Studies similar to those done for the normal gene will have to be done on the changed (mutated) copy of the gene, to see how the change in the gene changes the function of the protein that the gene makes. **SEE ALSO** COMPLEX TRAITS; GENE DISCOVERY; MAPPING; TWINS.

Jonathan L. Haines

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Internet Resource

Dolan DNA Learning Center. Cold Spring Harbor Laboratory. <<http://www.dnalc.org>>.

Human Genome Project

The **genome** represents the entire complement of DNA in a cell. The Human Genome Project is the determination of the entire nucleotide sequence of all 3 billion+ bases of DNA within the nucleus of a human cell. It is one of the greatest scientific undertakings in the history of mankind. The first draft of the human genome sequence was completed in the year 2001 and published simultaneously in the British journal *Nature* and the American journal *Science*.

The data obtained from sequencing the human genome promise to bring unprecedented scientific rewards in the discovery of disease-causing genes, in the design of new drugs, in understanding developmental processes and cancer, and in determining the origin and evolution of the human race. The Human Genome Project has also raised many social and ethical issues with regard to the use of such information.

Origins of the Human Genome Project

One could say that the Human Genome Project really began in 1953, when James Watson and Francis Crick deduced the molecular structure of DNA, the molecule of which the genome is made. (Watson and Crick were awarded the Nobel Prize for this work in 1962.) Since that time, scientists have wanted to know the complete sequence of a gene, and even dreamed that some day it would be possible to determine the complete sequence of all of the genes in any organism, including humans.

The original impetus for the Human Genome Project came almost a decade earlier, however, from the U.S. Department of Energy (DOE) shortly after World War II. The atomic bombs that were dropped on Hiroshima and Nagasaki, Japan, left many survivors who had been exposed to high levels of radiation. The survivors of the bomb were stigmatized in Japan. They were considered poor marriage prospects, because of the potential for carrying mutations, and the rest of Japanese society often ostracized them. In 1946 the famous geneticist and Nobel laureate Hermann J. Muller wrote in the *New York Times* that “if they could foresee the results [mutations among their descendants] 1,000 years from now . . . , they might consider themselves more fortunate if the bomb had killed them.”

Muller had firsthand experience with the devastating effects of radiation, having studied the biological effects of radiation on the fruit fly *Drosophila melanogaster*. He predicted similar results would follow from the human exposure to radiation. As a consequence, the Atomic Energy Commission of the DOE set up an Atomic Bomb Casualty Commission in 1947 to address the issue of potential **mutations** among the survivors. The problem they faced was how to experimentally determine such mutations. At that time there were no suitable methods to study the problem. Indeed, it would be many years before the appropriate technology was available.

During the 1970s molecular biologists developed techniques for the isolation and cloning of individual genes. Paul Berg was the first to create a recombinant DNA molecule in 1972, and within a few years gene cloning became a standard tool of the molecular biologist. Using cloning techniques, scientists could generate large quantities of a single gene, enabling researchers

genome the total genetic material in a cell or organism

mutations changes in DNA sequences



Table 1.

MODEL ORGANISMS SEQUENCED

Date sequenced ^a	Species	Total bases ^b
7/28/1995	<i>Haemophilis influenzae</i> (bacterium)	1,830,138
10/30/1995	<i>Mycoplasma genitalium</i> (bacterium)	580,073
5/29/1997	<i>Saccharomyces cerevisiae</i> (yeast)	12,069,247
9/5/1997	<i>Escherichia coli</i> (bacterium)	4,639,221
11/20/1997	<i>Bacillus subtilis</i> (bacterium)	4,214,814
12/31/1998	<i>Caenorhabditis elegans</i> (round worm)	97,283,371 99,167,964 ^c
3/24/2000	<i>Drosophila melanogaster</i> (fruit fly)	~137,000,000
12/14/2000	<i>Arabidopsis thaliana</i> (mustard plant)	~115,400,000
1/26/2001	<i>Oryza sativa</i> (rice)	~430,000,000
2/15/2001	<i>Homo sapiens</i> (human)	~3,200,000,000

^aFirst publication date.^bData excludes organelles or plasmids. These numbers should not be taken as absolute. Scientists are confirming the sequences; several laboratories were involved in the sequencing of a particular organism and have slightly different numbers; and there are some strain variations. Data were obtained from the (NCBI) Web site.^cThe first number was originally published, and the second is a correction as of June 2000.

to study its structure and function. In 1977 Drs. Walter Gilbert and Fred Sanger independently developed methods for the sequencing of DNA, for which they received the 1980 Nobel Prize along with Berg. Sanger's group in England was the first to completely sequence a genome, identifying all 5,386 bases of the bacterial virus $\phi\chi 174$.

Another technological breakthrough occurred in 1985, when the polymerase chain reaction method was developed by Dr. Kary Mullis and colleagues at Cetus Corp. This team devised a method whereby minute samples of DNA can be multiplied a billion-fold for analysis. This technique, which has many applications in diverse fields of biology, is one of the most important scientific breakthroughs in gene analysis. Mullis received the Nobel Prize for this work in 1993.

At this time, however, DNA sequencing was still done by hand. At best, a researcher could manually sequence only a few hundred bases per day. To be able to sequence the human genome, machines would be needed that could sequence a million or more bases per day. In 1986 Leroy Hood developed the first generation of automated DNA sequencers, thereby dramatically increasing the speed with which bases could be sequenced. Thus, by the mid-1980s the stage was set.

With these new techniques, molecular biologists now felt that it might be feasible to sequence the entire human genome. The first serious discussions came in June 1985, when Robert Sinsheimer, chancellor of the University of California at Santa Cruz, called a meeting of leading scientists to discuss the possibility of sequencing the human genome. Sinsheimer was inspired by the success of the Manhattan Project, which was the concerted effort of many physicists to develop atomic weapons during World War II. That project led to rapid development and a massive influx of funding for physicists. Sinsheimer wanted a "Manhattan Project" for molecular biology, to enhance and expand human genome research.

Meanwhile, the DOE continued to be interested in the problem of identifying mutations caused by radiation exposure. Led by associate director Charles DeLisi, the DOE became a strong supporter of the genome-mapping initiative, for it understood that sequencing the entire genome would provide the best way to analyze such mutations. Thus the DOE became the first federal agency to begin funding the Human Genome Project.

Mapping the human genome came to be called the “Holy Grail of Molecular Biology,” and many biologists were interested in the project. Most notable among them was Nobel laureate Gilbert who, through his interest, personality, and academic ties, developed enormous enthusiasm for the project. The initial goals set out for the Human Genome Project were three-fold: to develop genetic linkage maps; to create a physical map of ordered clones of DNA sequences; and to develop the capacity for large-scale sequencing, because faster and cheaper machines along with other great leaps in technology would be needed to get the job done.

In 1988 the National Institutes of Health (NIH) set up an Office of the Human Genome, and Watson agreed to head the project. It had an estimated budget of approximately \$3 billion, and 3 percent of the funding was devoted to the study of the social and ethical issues that would arise from the endeavor. A target date for completion of the project was set for September 30, 2005. By 1990 the Human Genome Project had received the additional endorsement of the National Academy of Sciences, the National Research Council, the DOE, the National Science Foundation, the U.S. Department of Agriculture, and the Howard Hughes Medical Institute. Sequencing of the human genome had now officially begun.

While sequencing the human genome was a primary goal, other sequencing projects were just as important. Many scientists established projects that sought to sequence several organisms of genetic, biochemical, or medical importance (see Table 1). These so-called model organisms, with their smaller genomes, would be useful in testing sequencing methodologies and for providing invaluable information that could be used to identify corresponding genes in the human genome. Sequence databases were established, and computer programs to search these databases were written.

Competition between the Public and Private Sectors

Dr. Craig Venter, a scientist at the NIH, felt that private companies could sequence genomes faster than publicly funded laboratories. For this reason he founded a biotechnology company called the Institute for Genomic Research (TIGR). In 1995 TIGR published the first completely sequenced genome, that of the bacterium *Haemophilus influenzae*. TIGR was soon joined by other biotechnology companies that competed directly with the publicly funded Human Genome Project.

Among these other biotech firms is Celera Genomics, founded in 1998 by Venter in conjunction with the Perkin-Elmer Corporation, manufacturer of the world’s fastest automatic DNA sequencers. Celera’s goal was to privately sequence the human genome in direct competition with the public efforts supported by the NIH and DOE and the governments of several foreign countries. Using 300 Perkin-Elmer automatic DNA sequencers along with one of the world’s most powerful supercomputers,



Celera sequenced the genomes of several model organisms with remarkable speed and, in April 2000, announced that it had a preliminary sequence of the human genome.

In order eventually to make a profit, these biotech companies were patenting DNA sequences and intended ultimately to charge clients, including researchers, for access to their databases. This issue of patenting had already caused controversy. Watson felt strongly that the sequence data flowing from the Human Genome Project should remain within the public domain, freely available to all. Meeting opposition to this view, he stepped down from his position as director of the NIH-sponsored project in 1992 and was succeeded by Francis Collins.

Other researchers shared Watson's view, and in 1996 the international consortium of publicly funded laboratories agreed at a meeting in Bermuda to release all data to GenBank, a genome database maintained by NIH. The agreement reached by these scientists came to be known as "The Bermuda Principles," and it mandated that sequence data would be posted on the Internet within 24 hours of acquisition. Because the information is freely available to the public, the sequences can not be patented. The dispute between Celera Genomics and the International Human Genome Consortium continues, as scientists now begin the task of searching the genome for valuable information.

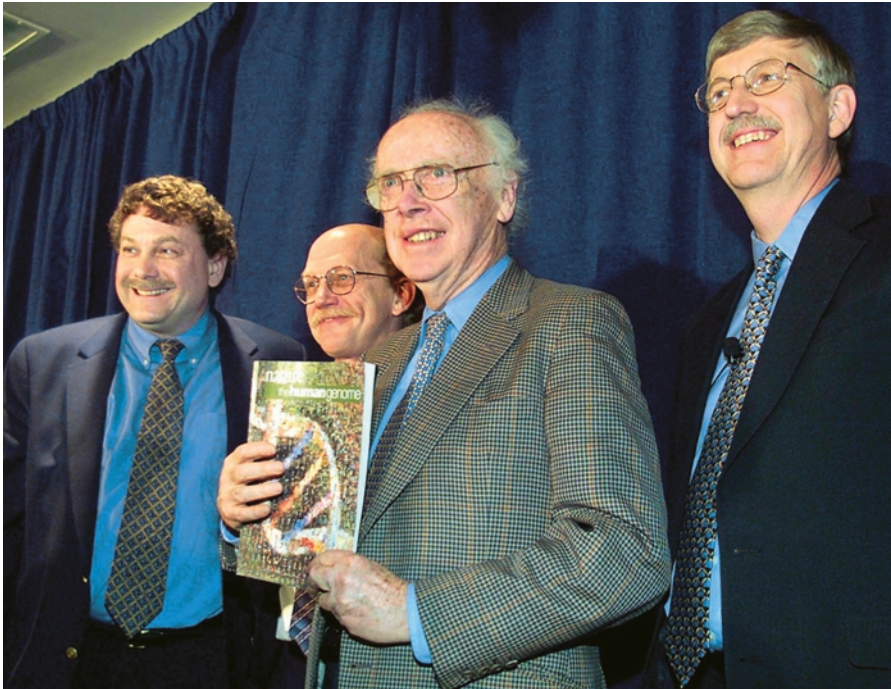
Progress in the Human Genome Project

Sequencing the human genome has led to some surprising results. For example, we once thought that highly evolved humans would need a great many genes to account for their complexity, and scientists originally estimated the number of human genes to be about 100,000. The draft of the human genome, however, indicates that humans may have only about 30,000 genes, far fewer than originally expected. Indeed, this is only about one-third more than the number of genes found in the lowly roundworm, *Caenorhabditis elegans* (approximately 20,000 genes), and roughly twice the number of genes in the fruit fly *Drosophila melanogaster* (approximately 14,000 genes). Subsequent estimates have placed the number of human genes closer to 70,000; the true number is unknown as of mid-2002. Scientists have learned that most of the genome does not code for proteins, but rather contains "junk DNA" of no known function. In fact, only a small percentage of human DNA actually encodes a gene.

The complete human genome consists of twenty-two pairs of chromosomes plus the X and Y sex chromosomes. On December 2, 1999, more than 100 scientists working together in laboratories in the United Kingdom, Japan, the United States, Canada, and Sweden announced the first completely sequenced human chromosome, chromosome 22, the smallest of the **autosomes**. To assure the accuracy of the sequence data, each segment must be sequenced at least ten times.

Thousands of scientists, working in more than 100 laboratories and 19 different countries around the world, have contributed to the Human Genome Project since its inception. Thanks to the development of later generations of high-speed automatic sequencers and supercomputers to handle the enormous amount of data generated, work on the project progressed well ahead of schedule and well under budget, a rare phenomenon in

autosomes chromo-
somes that are not sex
determining (not X or Y)



At a 2001 press conference, founder of the Human Genome Project, Dr. James Watson (with a copy of *Nature* in hand) announces the successful completion of human genome sequencing and poses with key supporters for photographers.

government-sponsored projects. In 2001 the first draft of the complete human genome was published. However, considerable work remained to be done, particularly in the sequence of regions of repetitive DNA.

Whose Genome Is It?

Although all humans share more than 99.99 percent of their genome sequences, each human is unique. Geneticists estimate that each person carries many mutations, perhaps hundreds or even thousands of them. Therefore, one of the major questions that has arisen in the Human Genome Project is “whose genome is it?” The final catalog of sequences, whenever it is complete, will have to take into account these individual variations, and ultimately there will be a “consensus sequence,” but it will represent no one specific individual.

A related issue arises from the distinct differences that scientists anticipate will occur among different populations. Which sequences should be considered “normal,” and which ones should be classed as “mutated”? The Human Genome Diversity Project was proposed in 1997 to catalog and study naturally occurring sequence variations among racial and geographic groups. This project never gained much support, however, because of the social and ethical ramifications to such a catalog. On the other hand, a Human Cancer Genome Anatomy Project was initiated to catalog all the genes that are expressed in cancer cells in order to aid in the detection and treatment of cancers. This project enjoyed much more support.

Patenting the Genome

From the outset there has been considerable debate among scientists, politicians, and entrepreneurs as to whether the human gene sequences can or

In 2002 Craig Venter announced that Celera had sequenced his personal genome, not a composite as originally claimed.



A Celera Genomics technician inputs DNA samples while the DNA sequencer machine searches and catalogs proteins.

should be patented. Indeed, this debate was the reason that Watson resigned as the first director of the NIH Human Genome Project program in 1992. Watson's position was opposed by many biotechnology companies, which hoped to recover the cost of their genome research and began patenting short segments of sequenced DNA without any idea as to their function. As of 2000, the U.S. Patent and Trademark Office (PTO) changed its policy, and began granting patents only to genes that have been identified, rather than just the random sequenced fragments. The data that flow from genome sequencing will be an invaluable scientific resource, particularly in the area of developing new medical treatments, but its use will be restricted if individual organizations can claim exclusive use rights to large segments of it. It is thus clear that debate on the patenting of genes will continue for years to come.

At present much of data from genome research are available to scientists and other interested parties. The data generated by participants in the Bermuda Principles agreement can be accessed on-line at the National Center for Biotechnology Information (NCBI) Web site, at <www.ncbi.nlm.nih.gov/genome/guide/human>. The International Human Genome Consortium Web site provides a current list of genome sites that offer links to most genome databases at <www.ensembl.org/genome/central>. Information about all the genomes that have been sequenced, as well as information on the sequencing of cancer genes, can be found on the Internet at <<http://www.ncbi.nlm.nih.gov>>.

Genomics and Proteomics

The Human Genome Project has given rise to new fields of research. One of these is **genomics**. This new field combines information science with molecular biology. It is resulting in the “mining of the genome” for valuable sequence data.

An even more recent development is the field of **proteomics**, the study of protein sequences. Research in this field is rapidly expanding, as protein sequences can be predicted from the gene sequence. The folding of the proteins (secondary and tertiary structures) can be predicted by computers, leading to a three-dimensional view of the protein encoded by a particular gene. Proteomics will be the next big challenge for genetics research. Indeed, Celera is already gearing up for massive protein sequencing.

genomics the study of gene sequences

proteomics the study of the full range of proteins expressed by a living cell

Ethical Issues

From the very beginning of the Human Genome Project, many from both the scientific and public sector have been concerned with ethical issues raised by the research. These issues include preserving the confidentiality of an individual's DNA information and avoiding the stigmatization of individuals who carry certain genes. Some fear that insurers will deny coverage for “preexisting” conditions to people carrying a gene that predisposes them to particular diseases, or that employers might start demanding genetic testing of job applicants.

There are also concerns that prenatal genetic testing could lead to genetic manipulation or a decision to abort based on undesirable traits disclosed by the tests. In addition, some raise concerns that a full knowledge of the human genome could raise profound psychological issues. For example, individuals who know that they carry detrimental genes may find the knowledge to be too great a burden to bear. All of these ethical issues will ultimately have to be addressed by society as a whole. SEE ALSO AUTOMATED SEQUENCER; BIOINFORMATICS; CANCER; CLONING GENES; GENOME; GENOMICS; GENOMICS INDUSTRY; MODEL ORGANISMS; MULLER, HERMANN; POLYMERASE CHAIN REACTION; POLYMORPHISMS; REPETITIVE DNA ELEMENTS; SEQUENCING DNA; WATSON, JAMES; YEAST.

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Human Immunodeficiency Virus *See HIV*

Huntington's Disease *See Triplet Repeat Disease*

Hybrid Superiority *See Heterozygote Advantage*

Immune System Genetics

The immune system is the set of cells and glands that protects the body from invasion and infection by viruses, bacteria, and other **pathogens**. The immune system must be able to recognize any foreign target, or antigen, of which there are potentially millions. Pathogenic organisms change over time, and new antigens evolve that must also be targeted. At the same time, the immune system must distinguish pathogenic antigens from the body's own tissues, attacking the former and sparing the latter. The key to the scope and specificity of the immune system response is in the genes that give rise to it.

Overview of the Immune System

The immune system includes several interacting components. Nonspecific immunity (protection against any invasion) is provided by the barriers of the skin and mucous membranes lining the lungs and gut. Additional non-specific defenses are provided by the inflammatory response and the complement proteins in the bloodstream. We shall not deal further with these defenses.

Specific immunity is the set of defenses mounted against a specific invader. It involves the action of three major types of cells: B cells, T cells, and macrophages. In broad, somewhat oversimplified terms, B cells make proteins called **antibodies** that attach to foreign antigens, serving as warning flags. T cells coordinate the immune attack and destroy virus-infected cells. Macrophages consume flagged antigens and clean up the debris from a T cell attack on infected cells.

An antibody binds to an invader when its shape fits some shape (the antigen) on the invader's surface. Any particular invader, such as a bacterial cell, may have dozens of such antigens.

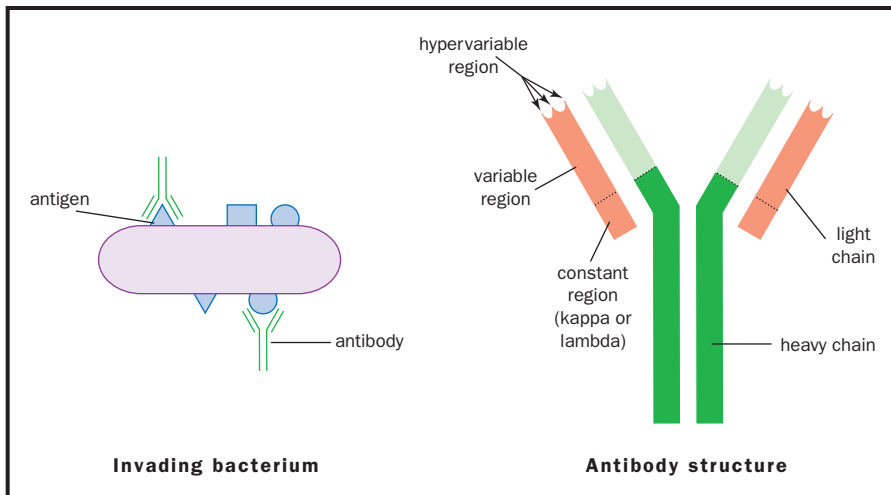
The Puzzle of Antibody Diversity

B cells are created in the bone marrow. Many millions of different B cells are made, each containing a unique gene for the specific antibody that it (and all its descendants) will make. A group of B cells with all its descen-



pathogens disease-causing organisms

antibodies immune-system proteins that bind to foreign molecules



Antibodies bind to antigens on the surface of foreign substances.

dants is called a clone. Thus, the antibody made by one B cell clone differs from that made by any other B cell clone. T cells develop along a slightly different pathway but also contain a unique protein, called the T cell receptor, which is coded for by a gene unique to that T cell clone.

Antibodies are proteins, and like all of the body's proteins, must be encoded by genes. However, the number of distinct antibodies each of us makes (many millions) is vastly greater than the total number of genes in our entire genome (30,000–70,000). How is all this diversity encoded? To understand the answer, it is helpful to look at the structure of an antibody.

Antibody Structure

The antibody is formed from four **polypeptides** that link up in the shape of a Y. There are two identical long heavy (H) chains and two identical short light (L) chains. The tips of each branch of the Y form a pocket, and it is here the antibody binds antigen. Thus, these twin pockets are called the antigen-binding regions of the antibody.

By comparing the amino acid sequences of antibodies from different B cell clones, several important features can be discovered. Light chains, for instance, have a constant region, with amino acid sequences that differ little from clone to clone, and a variable region, with sequences that differ considerably. The constant region comes in two different forms, termed “kappa” and “lambda.” The amino acid sequence of one kappa constant sequence differs little or not at all from clone to clone; similarly, all lambda constant sequences are essentially identical. The variable region does differ considerably between clones. The heavy chain also has a constant region (of which there are five forms) and a variable region.

The constant regions of all the chains are found toward the bottom of the Y, while the variable regions are found toward the tip. Furthermore, within each variable region, there are three hypervariable regions, whose five to ten amino acids differ even more than the other portions of the variable region. These hypervariable regions form the actual points of contact between antibody and antigen.

polypeptides chains of amino acids



Gene Segments Combine Randomly to Generate Diversity

The fundamental principle governing antibody generation is combinatorial diversity. A large number of genes are generated by choosing from among a smaller pool of differing gene segments and combining them in different ways. This process, known as **somatic** recombination, is similar in principle to constructing words. The alphabet's twenty-six letters can be combined to make 676 (26^2) two-letter words and almost 12 million five-letter words.

To understand the molecular details of somatic recombination, let us focus on the creation of a kappa-type light chain. The process is similar for lambda light chains and only marginally more involved for a heavy chain.

We noted that the light chain has both a variable and a constant region. There are forty gene segments that can code for the variable (V) region and a single segment that codes for the constant (C) region. In addition, there are five possible coding segments for the J region, a short region that is also present on light chains. All of these genes and segments are located in sequence on chromosome 2. Each V and J segment is flanked by special noncoding sequences that facilitate the next stage, in which specific segments are joined.

Somatic recombination begins when special recombining proteins randomly bring together the downstream end of one V segment and the upstream end of one J segment. They do this by attaching to the flanking sequences and bending the intervening DNA into a loop. The loop is cut out and degraded, and the remaining DNA is spliced together. The product is the mature antibody gene.

Note in the diagram on the right that the resulting gene may still have some extra upstream V segments. An ingenious mechanism prevents such segments from being transcribed to make messenger RNA, however.

Each V segment contains a **promoter**, the region to which RNA polymerase binds to start transcription. The promoter is inactive, though, until it is brought close to an “enhancer” region between the J and C segments. Thus, transcription will begin at the V segment closest to the enhancer, and only this one V segment is transcribed—the others are too far from the enhancer. The gene may also have extra downstream J segments and **intron** sequences between J and C. These are transcribed, but they are removed by RNA processing.

Other Sources of Diversity

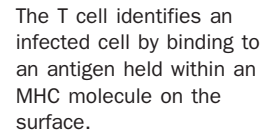
The random combination of V and J segments alone can produce millions of possible combinations. More diversity arises because the joining of V and J chains is done imprecisely, with the possible loss or gain of several nucleotides, resulting in added or deleted amino acids.

Remember also that each antibody includes both light and heavy chains. Heavy chains are produced by a similar combinatorial process, using a different, larger set of gene segments. The combination of a randomly produced light chain with a randomly produced heavy chain produces even more diversity. Finally, when a B cell multiplies in response to antigens, the

somatic nonreproductive; not an egg or sperm

promoter DNA sequence to which RNA polymerase binds to begin transcription

intron untranslated portion of a gene that interrupts coding regions



T Cell Receptors

When a T cell discovers a cell whose MHC proteins contain foreign antigens, it marks the cell for destruction. The T cell receptor interacts with antigens in much the same way as an antibody does, although the size of the antigen it recognizes is smaller. T cell receptors come in as many diverse forms as antibodies do, and, while the details differ, their diversity is generated in much the same way, with random recombination of gene segments.

The T cell-MHC interaction serves another, related function: It confirms that the cell is part of the self that the immune system should be protecting. Thus, MHCs serve as self-recognition markers. When a T cell recognizes foreign MHCs, as would occur in an organ transplant, it sets in motion an immune attack to reject the foreign tissue. Indeed, “histocompatibility” means compatibility of tissues, and these proteins control that process.